

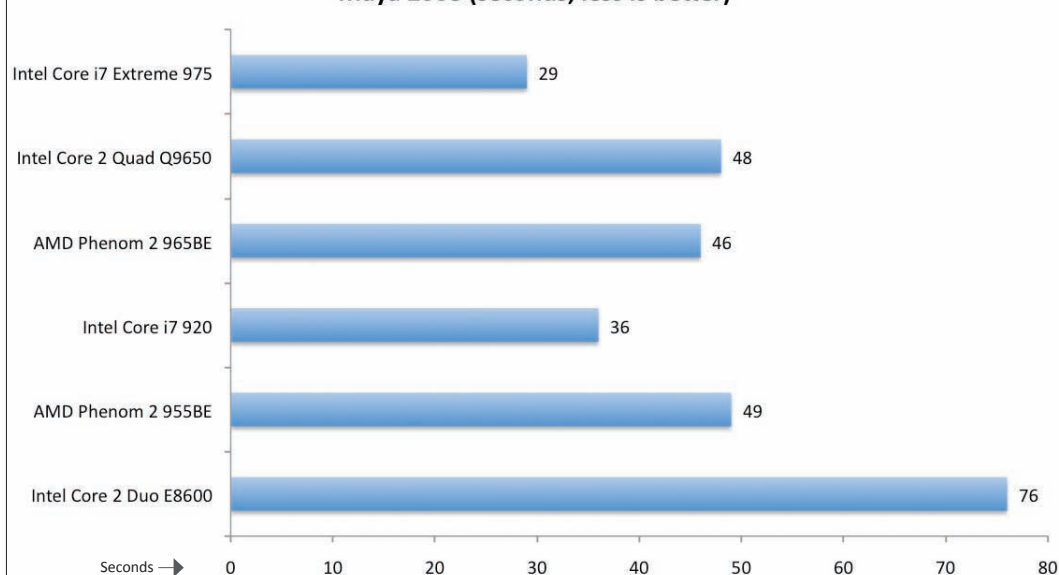
## CPU Test

Up To Rs. 5,000

| Brand                                                 | AMD           | Intel         | Intel                   | AMD              | AMD           | AMD              |
|-------------------------------------------------------|---------------|---------------|-------------------------|------------------|---------------|------------------|
| Model                                                 | Sempron 140   | Pentium E2200 | Pentium Dual Core E5200 | Athlon II X2 240 | Athlon 7850BE | Athlon II X2 250 |
| Price (Rs) *                                          | 2160          | 3500          | 3550                    | 3570             | 4110          | 4150             |
| Performance (Out of 100)                              | 37.71         | 28.55         | 34.77                   | 40.86            | 38.50         | 42.92            |
| <b>Features</b>                                       |               |               |                         |                  |               |                  |
| Socket Interface                                      | AM3/AM2+      | LGA 775       | LGA 775                 | AM3/AM2+         | AM2           | AM3/AM2+         |
| CPU Codename                                          | Sargas        | Allendale     | Wolfdale                | Regor            | Kuma          | Regor            |
| Core Frequency                                        | 2700 MHz      | 2200 MHz      | 2500 MHz                | 2800 MHz         | 2800 MHz      | 3000 MHz         |
| Number Of Cores / Hyperthreaded (Y/N)                 | 1 / N         | 2 / N         | 2 / N                   | 2 / N            | 2 / N         | 2 / N            |
| External Bus Frequency                                | 2000          | 800           | 800 MHz                 | 2000             | 2000          | 2000             |
| Fabrication Process (in Nanometers)                   | 45nm          | 65 nm         | 45nm                    | 45nm             | 65nm          | 45nm             |
| L1 Cache / L2 Cache                                   | 128 KB / 1 MB | 128 KB / 1 MB | 128 KB / 2 MB           | 256 KB / 2 MB    | 256 KB / 2 MB | 256 KB / 2 MB    |
| L3 Cache (if any)                                     | -             | -             | -                       | -                | -             | -                |
| <b>Performance</b>                                    |               |               |                         |                  |               |                  |
| <b>Synthetic benchmarks</b>                           |               |               |                         |                  |               |                  |
| 3D Mark Vantage (CPU Score)                           | 27442         | 26638         | 29749                   | 32762            | 31107         | 32980            |
| Wprime 2.00 (32 million, in sec) #                    | 54.537        | 36.77         | 30.624                  | 27.02            | 27.405        | 24.351           |
| ScienceMark 2.0 Cryptography (AES, Rijndael) in sec # | 9.707         | 14.079        | 12.279                  | 9.147            | 9.845         | 8.561            |
| ScienceMark 2.0 Membench (Memory bandwidth) MBps      | 9797.41       | 3891.86       | 4402.43                 | 9576.31          | 8057.53       | 9679.49          |
| Maxxon CineBench R10 rendering (all cores / one core) | NA / 2426     | 3971 / 2093   | 5118 / 2665             | 4909 / 2513      | 4878 / 2442   | 5237 / 2687      |
| PovRay 3.62 (Chess2 scene) (in sec) #                 | 182           | 256           | 218                     | 178              | 181           | 165              |
| Valve Multicore CPU Particle benchmark                | 23            | 37            | 49                      | 43               | 42            | 47               |
| <b>Real World Applications</b>                        |               |               |                         |                  |               |                  |
| Maya 2008 Rendering all cores / one core (in sec) #   | NA / 192      | 132 / 239     | 104 / 191               | 101 / 183        | 105 / 190     | 95 / 171         |
| WinRAR 3.9 (all cores / single core)                  | 673 / 666     | 843 / 466     | 978 / 530               | 1223 / 675       | 1094 / 597    | 1257 / 694       |
| DivX 7.0 Encoding (sec) #                             | 374.7         | 306.7         | 257.4                   | 233.3            | 248.2         | 221.7            |
| HandBrake x264 encoding (sec) #                       | 676.9         | 514.6         | 421.5                   | 385.7            | 392           | 358.6            |
| Doom 3 (FPS)                                          | 125.9         | 122.3         | 165.3                   | 207.8            | 190.4         | 222.6            |
| Crysis (CPU benchmark, FPS)                           | 74            | 67            | 79                      | 87               | 82            | 88               |
| COH: Tales of Valor (FPS)                             | 194.8         | 186.4         | 240.3                   | 295.4            | 288.2         | 311.6            |

\* Exclusive of all taxes # for these scores lower is better

Maya 2008 (seconds, less is better)



hotter than the rest – we reckon AMD could squeeze another 200 MHz out of the Deneb (codename for Phenom II) architecture but that's about it.

Our first surprise came in the Wprime 2.00 test where both the Phenom II 955BE and 965BE are faster than the mighty Q9650 and QX9650 chips – both potent 3.0 GHz, quad core parts; evidently AMD's higher clock speed coupled with the increase in its L3 cache from 2 MB on the first Phenom processors to 6 MB on these CPUs is enough to zoom ahead. Thanks to their slower, latency plagued, FSB, the Core 2 CPUs also have lower memory bandwidth scores in ScienceMark 2.0. The Phenom II 965BE is also faster than all Core 2 CPUs